

Final Project Report

Design and Implementation of an IT Infrastructure and Support Solution for Ubuntu Innovations (Pty) Ltd

A structured and professionally formatted deliverable for the Ubuntu Innovations infrastructure and support solution.

Client	Ubuntu Innovations (Pty) Ltd
Programme	CAPACITI Tech Career Accelerator
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Executive Summary

Ubuntu Innovations (Pty) Ltd is a growing Cape Town technology startup with 25 employees. This final report consolidates the capstone solution into one professional IT support and infrastructure document.

The solution covers business requirements, hardware and software recommendations, network architecture, IP addressing, user access, operating system administration, backup and recovery, cybersecurity, risk management, incident response, troubleshooting and career readiness materials.

Business Requirements Analysis

Department	Employees	Primary needs
Executive Management	3	Mobile access, reporting, approvals
Finance	4	Secure financial data, accounting tools, printers
Human Resources	3	Confidential employee records and controlled access
Sales & Marketing	7	Collaboration tools, CRM and wireless access
Software Development	8	Development tools, repositories and shared technical resources

- Secure wired and wireless connectivity
- Shared storage and printers
- Employee accounts and access control
- Endpoint security and patch management
- Backup, recovery and troubleshooting documentation

Hardware and Software Inventory

Hardware Recommendations

Asset	Qty	Recommended specification	Business justification
Workstation PCs	20	Intel Core i5, 16GB RAM, 512GB SSD, Windows 11 Pro / Ubuntu Linux dual boot	Reliable office workstations for Finance, HR, Sales and Development users
Corporate laptops	5	Intel Core i7, 16GB RAM, 512GB SSD, Wi-Fi 6, webcam and TPM security	Mobility for executives and hybrid work
Business router / firewall	1	VPN, IDS/IPS, VLAN support and traffic monitoring	Secure perimeter, VPN access and internet management
Managed switch	1	24-port Gigabit managed PoE+ switch	Central wired connectivity and VLAN support
Wireless access points	2	Dual-band Wi-Fi 6 access points	Secure wireless coverage for staff and guests
Network printers	2	Laser multifunction printers with network support	Shared printing and scanning
NAS device	1	8TB RAID storage with cloud sync	Shared file storage and local backup repository
UPS units	2	1500VA UPS with monitoring	Protects key infrastructure during power interruptions

Software and Licensing Plan

Area	Software / service	Licensing approach	Purpose
Company-wide	Windows 11 Pro / Ubuntu Linux, Microsoft 365 or Google Workspace, browser, PDF reader	Per-user or per-device licences	Productivity and standard workstation environment
Finance	Accounting software, spreadsheet tools, secure PDF tools	Named Finance licences	Financial processing and reporting
HR	HR records system, document management, secure storage	Named HR licences	Employee recordkeeping and confidential files
Sales & Marketing	CRM, design/campaign tools, communication tools	Department licences	Customer engagement and marketing operations
Development	VS Code, Git, GitHub, database clients, Docker optional	Open-source plus team licences	Software development and version control
Security	Endpoint protection, MFA, encryption and patching tools	Company-wide security licences	Endpoint and data protection

Network Design

The network design connects the ISP to a business-grade firewall router, then to a managed switch that supports wired user devices, wireless access points, network printers and NAS storage. Guest Wi-Fi should remain isolated from internal resources.

IP Addressing Plan

Device group	Subnet	Example range	Purpose
Servers / NAS	192.168.10.0/24	192.168.10.10 - 192.168.10.50	Infrastructure and shared services
User devices	192.168.20.0/24	192.168.20.10 - 192.168.20.200	Desktops and laptops
Printers	192.168.30.0/24	192.168.30.10 - 192.168.30.30	Static printer addresses
Corporate Wi-Fi	192.168.40.0/24	192.168.40.10 - 192.168.40.200	Staff wireless access
Guest Wi-Fi	192.168.50.0/24	192.168.50.10 - 192.168.50.200	Isolated guest internet access

User and Permissions Matrix

Department	Security group	Folder access	Permission level
Management	mgmt_grp	/Company/Management and oversight folders	Full access where business approved
Finance	finance_grp	/Company/Finance	Modify finance content only
Human Resources	hr_grp	/Company/HR	Modify HR content only
Sales & Marketing	sales_grp	/Company/Sales	Modify sales content only
Software Development	dev_grp	/Company/Development	Modify development content only
All staff	all_staff	/Company/Public	Read/write public shared content

Operating System Administration

Ubuntu Linux administration tasks include navigating the file system, managing files and permissions, monitoring processes and installing packages through the command line.

Command	Example	Purpose
pwd	pwd	Show current working directory
ls	ls -la	List files and permissions
cd	cd /Company/Public	Change directory
mkdir	mkdir Reports	Create a folder
touch	touch notes.txt	Create an empty file
chmod	chmod 770 Finance	Change file/folder permissions
chown	chown user:group file	Change owner and group
ps	ps aux	View running processes
top	top	Monitor live system activity
apt install	sudo apt install package	Install software on Ubuntu

Backup and Recovery Plan

Backup type	Frequency	Storage location	Purpose
Incremental backup	Daily	NAS and encrypted cloud copy	Protects recent changes with lower storage use
Full backup	Weekly	NAS storage	Complete weekly recovery point
Archive backup	Monthly	Offsite cloud storage	Longer-term retention and compliance
Restore testing	Monthly sample restore	Test environment	Confirms backups can actually be restored

- Daily incremental backups protect recent changes.
- Weekly full backups provide a complete restore point.
- Monthly archive backups support long-term retention.
- Offsite cloud backup protects against site-level loss.

Cybersecurity Policy

Control	Requirement	Business reason
Password complexity	Minimum length, complexity and no password sharing	Reduces credential compromise
Multi-factor authentication	Required for email, cloud services and admin accounts	Protects accounts if passwords are exposed
Device encryption	Enable encryption on laptops and sensitive systems	Protects data if devices are lost or stolen
Endpoint protection	Antivirus/EDR on all endpoints	Detects malware and malicious behaviour
Patch management	Monthly patch cycle with urgent security exceptions	Reduces known vulnerabilities
Acceptable use	Business systems used responsibly and legally	Sets clear user expectations

Risk Assessment

Risk	Likelihood	Impact	Mitigation
Phishing	High	High	Security awareness training, MFA and reporting process
Ransomware	Medium	High	EDR, patching, tested backups and least privilege
Hardware failure	Medium	Medium	UPS, warranty support and backup hardware planning
Power outage	Medium	High	UPS protection and safe shutdown procedure
Data loss	Medium	High	Daily backups, offsite storage and restore testing
Unauthorised access	Medium	High	Access reviews, MFA and role-based permissions

Incident Response Plan

Stage	Action	Owner / notes
Detection	Identify suspicious activity, outage or user report	User / IT Support
Reporting	Log ticket and notify appropriate escalation contact	IT Support
Containment	Isolate affected device, account or network segment	IT Support / Security
Eradication	Remove malware, disable compromised access or correct root cause	IT Support / Security
Recovery	Restore systems from backups and confirm normal operation	IT Support
Lessons learned	Document findings and update controls/training	IT Lead / Management

Troubleshooting Guide

Scenario	Symptoms	Likely causes	Resolution approach
Desktop not powering on	No lights, no fan sound, blank display	Power cable, wall socket, PSU or internal hardware fault	Check power source, cables and outlet; test known-good cable; escalate hardware repair if needed
No internet connectivity	Cannot browse, email fails, Wi-Fi disconnected	Adapter issue, DHCP/DNS problem, access point or ISP issue	Check scope, renew IP, test gateway/DNS, inspect network devices and escalate ISP if needed
Printer offline	Print jobs stuck, device shows offline	Network issue, incorrect IP, print queue or driver problem	Check printer network status, restart queue, confirm IP, update driver or re-add printer

Career Readiness Materials

The career readiness deliverables include a personalised IT Support CV, tailored cover letter, LinkedIn profile summary and complete interview preparation guide for Jodie Solomons.

- The CV highlights IT Support training, software development experience, troubleshooting skills and technical tools.
- The cover letter is tailored for a Junior IT Support Specialist role.
- The LinkedIn summary positions the candidate for IT Support, Technical Support, Service Desk and Junior System Administration roles.
- The interview guide includes ten complete sample answers.

Conclusion and Recommendations

The proposed solution provides Ubuntu Innovations with a secure, scalable and supportable IT environment. It balances practical small-business needs with professional controls suitable for growth.

- Implement MFA across business-critical systems.
- Test backups regularly and document restore results.
- Review access permissions at least annually.
- Run quarterly security awareness training.
- Maintain a structured patch management schedule.
- Consider Active Directory or cloud identity integration as the business grows.

References

CAPACITI Tech Career Accelerator. Four Week Capstone Project Guide.

Google IT Support Professional Certificate course modules.

Project documentation created for the Ubuntu Innovations IT Support Capstone.

Appendices

Appendix A: Network diagrams. Appendix B: Hardware and software inventories. Appendix C: User and permissions matrix. Appendix D: Risk and incident response documentation. Appendix E: Career readiness documents.